# Computer session BE 1 (4 hours)

ENSE3 ASI 2A

2024-2025

## General introduction

For each computer session, you will work in a group of two and you have to submit an electronic report in pdf format. The report must be submitted to the dedicated folder on Chamilo. The group number and members' names must clearly mentioned. For figures, **no screen shots are allowed.** Rather use the figure environment of Matlab. Don't forget the figures titles and legends. Equations must be typeset using the equation environmental of Latex or word. Essential Matlab code can be included.

## Small questions

- (**Easy**) Plot the contour lines of the Rosenbrock function given by :

  $$f(x) = b(x_2 - x_1^2)^2 + (a - x_1)^2$$

  for $a = 1$ and $b = 4$. Find, using Matlab, its gradient at the point $x = [1\ 2]^T$.

- (**Moderate**) Consider this nonlinear function :

  $$f(x; y) = \frac{1}{2}(x-1)^2 + \frac{1}{2}(10(y - x^2))^2 + \frac{1}{2}y^2$$

  Derive by hand its gradient and Hessian matrix and then find the same result using the symbolic toolbox of Matlab.

- (**Moderate**) Consider the following Lotka–Volterra model for a predator-prey system, known as the

  $$\begin{aligned} \frac{dx_1}{dt} &= b_r x_1 - a x_1 x_2 \\ \frac{dx_2}{dt} &= b x_1 x_2 - d_f x_2 \end{aligned}$$

  Here $x_1$ represents the population of the prey and $x_2$ represents the population of the predator. Using Matlab, draw the phase portrait (i.e. $x_2$ versus $x_1$) using these parameters $b_r = d_f = a = b = 1$.

- (**Moderate**) Consider a set of data points:

  $$x = [1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10]$$

  $$y_1 = [0.2\ 0.5\ 0.3\ 0.7\ 1.0\ 1.5\ 1.8\ 2.0\ 2.3\ 2.2]$$

  We want to fit a curve (line) to this data set, by minimizing respectively the Maximum Error ($l_\infty$), Mean Absolute Error ($l_1$), and Least-squares Error ($l_2$). Compare your "line fits" for the three different error metrics. For some reason, we have outliers in the data as you will notice below ($y_2$), do we have a change in the predictions? if yes, explain why by comparing your plots in both cases (without and with outliers)?

  $$y_2 = [0.2\ 0.5\ 0.3\ 3.5\ 1.0\ 1.5\ 1.8\ 2.0\ 2.3\ 2.2]$$

  Hint: you can use 'fminsearch' and 'polyval' matlab functions.

# Problem 1

(**Moderate**) In Brazil, a common joke between undergrad students (usually during exams) is that "maybe I should give up and go sell my art on the beach". This was the case of Javier.

He spends his days on the beach, where he showcases his products, and while not selling, he can work on more of them. It's a simple life. However, while working on his art, Javier's mind wanders into his optimization class. He realizes that he could some use some of the basics of that class to improve his production. Javier has 3 products:

- Necklaces
- Bracelets
- Earrings

To create these products, he uses 3 materials:

- Beads
- String
- Wire

Let's simplify, and assume all necklaces are equal, all bracelets are equal, etc. For each product, you have a different use of materials, a different selling price, and a different time to assemble them, as described in the table 1: Additionally, there are costs for the materials (table 2). Finally, Javier assumes

| | beads (u) | string (m) | wire (m) | price (€) | hours (h) |
|---|---|---|---|---|---|
| bracelet | 15.0 | 0.6 | 0.15 | 25.0 | 2.5 |
| necklace | 80.0 | 2.5 | 0.2 | 45.0 | 4.0 |
| earring | 4.0 | 0.1 | 0.15 | 10.0 | 1.5 |

Table 1: Some information on the problem.

| | beads (u) | string (m) | wire (m) |
|---|---|---|---|
| cost (€) | 0.25 | 0.7 | 1.5 |

Table 2: Cost

that having variety is good for business, so he want at least some amount of each product (table 3 ).

| | bracelet | necklace | earring |
|---|---|---|---|
| minimum production | 2 | 2 | 2 |

Table 3: Minimum production limits.

Javier will work for 16 hours and he wants to maximize his profit while respecting this time availability and the minimum production limits. Give the detailed formulation of this nice optimization problem (cost and constraints).